The Ising perceptron at nonzero margin: certified bounds along the capacity curve

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July 2026

Abstract

The storage capacity of the Ising perceptron at margin κ is predicted by a replica-symmetric variational problem whose zero-margin instance, the Krauth–Mézard constant $\alpha_\star \approx 0.833$, was recently settled: Ding and Sun proved the lower bound, Huang proved the upper bound at general margin conditionally on four numerical hypotheses, and the zero-margin instances of those hypotheses have been verified in interval arithmetic. This paper moves the verified upper bound off the point $\kappa = 0$. Two elementary identities make the general-margin system as tractable as the zero-margin one: the replica-symmetric free energy is stationary in both order parameters exactly at the fixed point, so on each certified branch rectangle the threshold $\alpha_\star(\kappa)$ is the unique root of a strictly decreasing function; and $\partial R/\partial q$ collapses to a one-term integral with positive integrand, a form of the general-margin monotonicity of Ding and Sun that interval arithmetic can consume directly. Building on these we verify, in ball arithmetic over sub-slabs of the margin axis, Huang's fixed-point, AMP, and local-concavity conditions on the strip $\kappa \in [-0.65, 0.19]$ (1680 contiguous slabs across two uniqueness lanes, the second resting on Nakajima's saddle theorem under a certified premise), with $\alpha_\star(\kappa)$ located to width $2 \cdot 10^{-6}$ at five margins and the local maximality of the first-moment functional at its degenerate maximizer certified at six. The remaining hypothesis, the global nonpositivity of Huang's two-variable rate function, we verify at all five margins by porting the zero-margin certificate architecture to located parameter boxes: the maximizer is kept symbolic through the base-point identity $\nabla\Phi(1, 0) = (\psi_0(1 - q_0), q_0)$, the dual localization is evaluated through a correlated Bregman integrand that collapses algebraically, and the far-tail kernels are stabilized by classical Mills-ratio bounds. At each of the five margins all four hypotheses hold, so Huang's theorem gives unconditional capacity upper bounds, for instance $\alpha_\star(0.05) \in [0.781073068, 0.781074776]$ and $\alpha_\star(-0.45) \in [1.557652103, 1.557654133]$. The verification code, certificates, and replay instructions are at `https://github.com/yspennstate/ising-perceptron-margin`. On the lower-bound side, open at every nonzero margin, we prove that unconditioned second moments fail at every margin (with a closure over nonnegative weights for $\kappa \geq 0$), derive capacity lower bounds for all $\kappa < 0$ by embedding a symmetric perceptron, and verify in floating point that the general-margin form of Ding–Sun's lower-bound condition holds at all five certified margins.

1 Introduction

Store $M = \alpha N$ random patterns in a linear classifier with binary weights: for i.i.d. standard Gaussian vectors $g_1, \dots, g_M$ in $\mathbb{R}^N$ and a margin $\kappa \in \mathbb{R}$, ask whether some $w \in \{-1, +1\}^N$

satisfies

$$\frac{\langle g_a, w\rangle}{\sqrt{N}} \;\geq\; \kappa \qquad \text{for all } a \leq M.$$

Krauth and Mézard [1] computed the replica-symmetric prediction for the largest such α and conjectured it sharp; at $\kappa = 0$ their constant is $\alpha_\star = 0.8330786\ldots$ Three decades later the zero-margin conjecture was resolved — Ding and Sun [3] proved the lower bound and Huang [4] the upper bound — but both proofs are conditional on numerical hypotheses about explicit low-dimensional variational problems, and the recent interval-arithmetic verification of those hypotheses [6] is a zero-margin verification: the certified statement stops at $\kappa = 0$.

Huang's upper-bound theorem, however, is stated at every margin. For each κ it bounds the capacity by $\alpha_\star(\kappa)$, the root of the replica-symmetric free energy along its fixed-point branch, conditionally on four numerical hypotheses at that margin: a fixed-point condition (existence, uniqueness, and the free-energy crossing), a condition on the associated approximate-message-passing iteration, a local-concavity condition at a witness point, and a global nonpositivity condition for a two-variable rate function $\mathcal{S}_\star$. At $\kappa = 0$ these are exactly the hypotheses closed in [6]; at $\kappa \neq 0$ they were open. The general-κ curve has been predicted before by non-rigorous means — Stojnic's fully-lifted random duality theory [12] reproduces the same replica-symmetric values across margins — so what is new here is not the numerical value but its rigorous certification, which had stopped at $\kappa = 0$.

This paper is a certified study of the whole margin axis. Its contributions are of three kinds.

First, two identities (Section 3) that make the general-margin system as well behaved as the zero-margin one. The replica-symmetric functional G is stationary in both order parameters exactly on the fixed-point branch, at every (α, κ); by the envelope theorem, along a differentiable fixed-point branch, $G_\star'(\alpha) = \mathbb{E} \log \Psi(u) < 0$ strictly, so on every certified rectangle — where the contraction makes the branch exist, stay unique, and vary smoothly — $\alpha_\star(\kappa)$ is the unique root of a strictly decreasing function. And the derivative $\partial R/\partial q$ of the constraint-side fixed-point map collapses, after an integration by parts, to a one-term integral with positive integrand — the general-margin form of Ding–Sun's monotonicity lemma, and the identity that makes a tight interval enclosure of the contraction possible at all.

Second, ball-arithmetic certificates (Sections 5 and 6) discharging the fixed-point, AMP, and local-concavity hypotheses on the strip $\kappa \in [-0.65, 0.19]$, covered by 1680 contiguous slabs of width $5 \cdot 10^{-4}$. Uniqueness of the fixed point is carried by a per-slab contraction bound up to $\kappa = 0.10$ and, past the reach of that rectangle bound (a method wall, not a mathematical one — we measure the distinction precisely), by Nakajima's saddle-uniqueness theorem [5] under a premise certified in ball arithmetic. At five margins spanning the strip a certified bisection locates $\alpha_\star(\kappa)$ to width about $2 \cdot 10^{-6}$, and at six margins the Hessian of the first-moment functional at its degenerate maximizer is certified negative definite — the local ingredient of the global hypothesis.

Third, a certified verification of the global two-variable hypothesis at the margin $\kappa = 0.05$ (Sections 7 and 8). The zero-margin certificate architecture of [6] — a compact moment-coordinate sweep plus ray concavity on a star around the degenerate maximizer — does not transfer as-is, because at $\kappa \neq 0$ the fixed-point parameters (q_0, ψ_0) are known only as located boxes of width up to 10^{-4}, four orders wider than the zero-margin rectangle, and that width drowns every value-level enclosure the original design relies on. The port that works keeps the maximizer symbolic through the base-point identity $\nabla\Phi(1, 0) = (\psi_0(1 - q_0),\, q_0)$, evaluates the dual localization through a correlated Bregman integrand that collapses algebraically to $\log(\cosh\Delta + M\sinh\Delta) - M\Delta$, and stabilizes the far-tail second-derivative kernels by classical Mills-ratio bounds ($E' \in (0,1)$, $E'' \in (0, E-V)$). With those in place all four pieces of the zero-margin architecture certify at $\kappa = 0.05$, and Huang's theorem yields an unconditional upper bound at a nonzero margin.

Everything here concerns the upper bound. Ding–Sun's lower-bound argument is proved at $\kappa = 0$ only, so no statement in this paper should be read as an exact threshold at $\kappa \neq 0$ (Section 14).

Relation to the zero-margin verification

The certified primitives (ball-arithmetic special functions, rigorous quadrature, exact serialization) are those of [6], and at $\kappa = 0$ every pipeline in this paper reproduces the certified zero-margin values — the fixed-point rectangle, Huang's AMP and local-concavity constants, and his Hessian intervals — which we use throughout as anchors. The mathematical content specific to this paper is the pair of identities of Section 3, the general-margin Hessian formulas of Section 6, and the stability layer of Section 8; the engineering content is the located-box certificate architecture, which is what a margin interval demands once no analytically certified parameter rectangle exists.

2 The system and the four hypotheses

Let φ and Ψ denote the standard Gaussian density and upper tail, $M = \varphi/\Psi$ the inverse Mills ratio (an increasing function; $M' = M\,(M-u) \in (0,1)$), and $Z \sim N(0,1)$. Fix $\kappa \in \mathbb{R}$ and write

$$u(z; q, \kappa) = \frac{\kappa - \sqrt{q}\,z}{\sqrt{1-q}}, \qquad 0 < q < 1.$$

Following Ding–Sun and Huang, the replica-symmetric system is

$$P(\psi) = \mathbb{E}\big[\tanh^2(\sqrt{\psi}\,Z)\big], \qquad R(q,\alpha) = \frac{\alpha}{1-q}\,\mathbb{E}\big[M(u(Z))^2\big], \tag{1}$$

$$G(\alpha, q, \psi) = -\frac{\psi(1-q)}{2} + \mathbb{E}\log\big(2\cosh(\sqrt{\psi}\,Z)\big) + \alpha\,\mathbb{E}\log\Psi\big(u(Z)\big), \tag{2}$$

with the fixed-point branch $q = P(\psi)$, $\psi = R(q,\alpha)$ and the threshold $\alpha_\star(\kappa)$ defined as the zero of $G_\star(\alpha) = G(\alpha, q_\star(\alpha), \psi_\star(\alpha))$. At $\kappa = 0$, where $u = -\gamma z$ in law with $\gamma = \sqrt{q/(1-q)}$, this is the certified zero-margin system. (Huang's displayed free energy carries a sign typo on the α term; the fixed point requires the plus sign, which is what Ding–Sun use and what reproduces the certified rectangle.)

Huang's theorem bounds the capacity at margin κ by $\alpha_\star(\kappa)$ subject to four numerical hypotheses, which we state in the form his proof consumes.

Condition 1 (fixed point)**.** *On a stated rectangle, $q \mapsto P(R(q,\alpha))$ has a unique fixed point for every α in a stated interval, with $\sup_{q\in(q_{\mathrm{lb}},q_{\mathrm{ub}})}(P\circ R_\alpha)'(q) < 1$ uniformly on the rectangle, and $G_\star$ crosses zero inside that interval. The rectangle is existentially quantified: Huang's own $\kappa = 0$ instantiation (his Proposition 3.2) takes an interval of width 10^{-9} about the fixed point.*

Condition 2 (AMP)**.** $\alpha\,\mathbb{E}[\operatorname{sech}^4(\sqrt{\psi_0}Z)]\,\mathbb{E}[M'(u(Z;q_0,\kappa))^2]/(1-q_0)^2 < 1$ *at the fixed point, with $u(z;q,\kappa) = (\kappa - \sqrt{q}\,z)/\sqrt{1-q}$ as above.*

Condition 3 (local concavity). *$\lambda(z) < 0$ for some $z > -1$, where λ is the explicit function recalled in Section 5; a single witness suffices.*

Condition 4 (two-variable function). *$\mathcal{S}_\star(\lambda_1, \lambda_2) \le 0$ for all $(\lambda_1, \lambda_2) \in \mathbb{R}^2$, where $\mathcal{S}_\star$ is the rate function recalled in Section 7; equality holds at $(1, 0)$.*

3 Two identities

Lemma 1 (stationarity). *For all $\alpha > 0$, $\kappa \in \mathbb{R}$, $\psi > 0$ and $0 < q < 1$,*

$$\frac{\partial G}{\partial \psi} = \frac{q - P(\psi)}{2}, \qquad \frac{\partial G}{\partial q} = \frac{\psi - R(q, \alpha)}{2}.$$

Proof. The first identity is Gaussian integration by parts on the $\log\cosh$ term. For the second, differentiating u in q gives

$$u_q = \frac{1}{2\sqrt{1-q}}\Big(-\frac{z}{\sqrt{q}} + \frac{u}{\sqrt{1-q}}\Big), \qquad \frac{\partial}{\partial q}\,\mathbb{E}\log\Psi(u) = -\mathbb{E}\big[M(u)\,u_q\big].$$

Stein's lemma applied to the z part, $\mathbb{E}[zM(u)] = -\sqrt{q/(1-q)}\;\mathbb{E}[M'(u)]$, together with the Mills identity $M'(u) = M(u)^2 - uM(u)$, collapses the two pieces:

$$\mathbb{E}[M(u)\,u_q] = \frac{\mathbb{E}[M'(u)] + \mathbb{E}[u\,M(u)]}{2(1-q)} = \frac{\mathbb{E}[M(u)^2]}{2(1-q)} = \frac{R(q,\alpha)}{2\alpha}.$$

Hence $\partial_q G = \psi/2 - \alpha \cdot R/(2\alpha) = (\psi - R)/2$. ☐

Both derivatives vanish exactly on the fixed-point branch, so the envelope theorem gives $G_\star'(\alpha) = \mathbb{E}\log\Psi(u) < 0$ strictly, for every κ: the root $\alpha_\star(\kappa)$ is unique on the branch. This is the general-margin form of the monotonicity clause in Condition 1, and it is also what turns the located fixed-point boxes of Section 5 into certified $\alpha_\star$ intervals: a certified sign of G at a located box is a certified side of $\alpha_\star$.

Lemma 2 (derivative of R). *For all $\alpha > 0$, $\kappa \in \mathbb{R}$ and $0 < q < 1$,*

$$\frac{\partial R}{\partial q} = \frac{\alpha}{(1-q)^2}\,\mathbb{E}\Big[M'(u)^2 + 2\,M(u)^2 M'(u)\Big] \;>\; 0.$$

Proof. Differentiating (1) in q produces $R/(1-q) + \frac{\alpha}{1-q}\mathbb{E}[2MM'u_q]$. Splitting u_q as in Lemma 1 and applying Stein's lemma to $\mathbb{E}[z\,MM'(u)]$ (with $(MM')' = M'^2 + MM''$ and $M'' = M'(M - u) + MM' - M$) turns the cross term into

$$\mathbb{E}[2MM'u_q] = \frac{\mathbb{E}[M'^2 + 2M^2M' - M^2]}{1-q},$$

and the $-\mathbb{E}[M^2]/(1-q)^2$ piece cancels $R/(1-q)$ exactly. Positivity is immediate since $M' \in (0, 1)$. ☐

Lemma 2 replaces Ding–Sun's Lemma 7.2 (monotonicity of R, stated at $\kappa = 0$) at every margin. Its one-term form matters twice over: the integrand is a product of quantities with classical bounds, so its interval enclosure is tight enough for the per-slab contraction certificates; and positivity makes $q \mapsto P(R(q))$ increasing, which is what the contraction-free location of Section 5 bisects on. Both identities were checked against high-precision finite differences (agreement at the 10^{-15} level along the whole continuation) and symbolically.

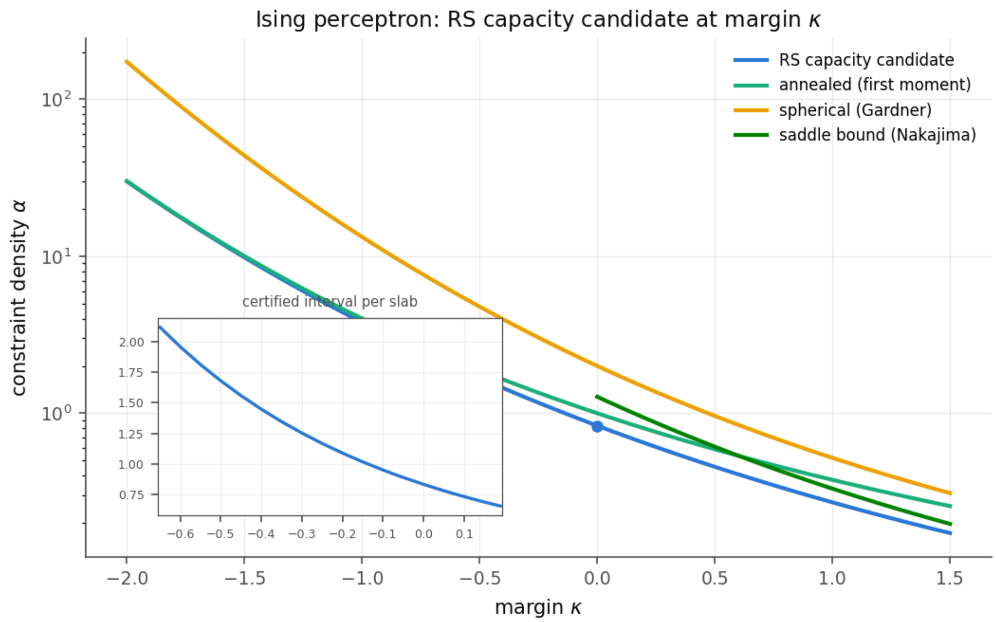

Figure 1: The replica-symmetric threshold $\alpha_\star(\kappa)$ (warm-started continuation from the certified value at $\kappa = 0$), the annealed bound $\log 2/(-\log \Psi(\kappa))$, the Gardner spherical capacity [2], and Nakajima's saddle-existence bound $\alpha_c(\kappa) = 2/(\pi\,\mathbb{E}[(\kappa - Z)_+^2])$ for $\kappa \geq 0$. The inset shows the certified per-slab $\alpha_\star$ intervals on the strip.

4 The capacity curve and the condition map

Before certifying anything we map the terrain in high-precision (nonrigorous) arithmetic: a warm-started continuation of the fixed point along $\kappa \in [-2.0, 1.5]$, recording at each of 71 points the threshold, the fixed point, the contraction $(P \circ R_\alpha)'(q_\star)$, the AMP product, the local-concavity minimum λ_0, and the stationarity residuals (which sit at the finite-difference noise floor, confirming Lemma 1 pointwise). Selected values:

κ	$\alpha_\star$	$q_\star$	$(P \circ R)'$	AMP	λ_0
-2.0	29.9921	0.11863	0.387	0.440	-0.147
-1.0	3.79463	0.36519	0.583	0.607	-0.099
-0.5	1.67884	0.47377	0.666	0.589	-0.132
0	0.83308	0.56395	0.744	0.544	-0.191
0.5	0.45566	0.63841	0.810	0.489	-0.271
1.0	0.27070	0.69988	0.863	0.433	-0.359
1.5	0.17245	0.75049	0.903	0.381	-0.442

All three per-point conditions hold at every point computed. The contraction is the only quantity trending toward failure, and only slowly; λ_0 approaches zero nearest $\kappa \approx -1.3$ (about -0.094) and then strengthens again — an early extrapolation from $[-0.55, 0]$ suggested a crossing near -1.5, and the measurement refutes it, a small lesson in not projecting fitted decay past its support. On the positive branch $\alpha_\star(\kappa)$ stays below Nakajima's bound at every grid point (worst ratio 0.874 at $\kappa = 1.5$). Toward $\kappa = -2$, the lower edge of the computed range, the threshold approaches the annealed bound from below, as it must where the constraints trivialize. The optimal tilt of the two-variable functional at its maximizer matches $\sqrt{1-q_0}$ to eight digits at every κ checked, confirming numerically Huang's identity for the tilt (stated by him at general margin) along the curve — this observation later becomes exact and load-bearing (Section 8).

5 Certificates on a strip

For κ in sub-slabs of width $5 \cdot 10^{-4}$ covering $[-0.65, 0.19]$, with per-slab parameter rectangles seeded by the continuation (the certificate is the proof; a too-tight seed simply fails and is regenerated wider), we verify Conditions 1–3 in ball arithmetic (`python-flint`, 80-bit working precision, every input a ball covering the whole slab).

5.1 Numerical architecture

Three implementation laws carry the whole computation; each was learned from a measured failure, and the failed artifacts are kept in the repository as evidence.

First, adaptive complex quadrature cannot resolve enclosure width that comes from parameter balls — subdividing the integration variable does not shrink it, and the integrator exhausts any budget — so every wide-parameter integral is a real-cell Riemann sum with exact per-cell Gaussian masses. Second, the ball evaluation of φ/Ψ fails on wide arguments in the steep region (the enclosure of Ψ crosses zero), while the ratio is monotone; all Mills-family factors are therefore enclosed by two thin endpoint evaluations, and the sign-critical sums use the mean-value cell rule (midpoint value plus a derivative correction, $O(h^2)$ against the plain hull's $O(h)$). Third, expectations that nearly cancel must be evaluated as one correlated integrand, never as separate sums whose parameter widths add. For Condition 3 this is the identity (with $t = M'$, $A = 1 - q_0$, and $b = m(\hat{z}) - A$ for $m(\hat{z}) = \mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0}Z))^{-1}]$)

$$\lambda(\hat{z}) = \hat{z} + \alpha\,\mathbb{E}\left[\frac{b\,t^2}{A(A+bt)}\right],$$

whose integrand is monotone in t for fixed (m, A), regular on all of $t \in [0,1]$, and bounded by $|b|/(Am)$; the separated form's enclosure crossed zero on the $[-0.45, -0.35]$ batch while the true value stays near -0.14. The identity, the monotonicity, and the bound are verified symbolically.

5.2 The contraction lane

On each slab the checks are: (i) the contraction $\sup_q (P \circ R_\alpha)' \le \sup_\psi P' \cdot \sup_q |\partial R/\partial q| < 1$ on the slab rectangle, via Lemma 2; (ii) existence — the endpoints of the q interval map strictly inward under $P \circ R_\alpha$ on each of eight α sub-intervals; (iii) the crossing — at the two thin α endpoints a certified Picard iteration locates the fixed point (the invariant is that the image of any set containing the fixed point contains it) and the sign of G resolves through a mean-value form that keeps the stationarity cancellations of Lemma 1 explicit, $G(\alpha_{\mathrm{lb}}) > 0 > G(\alpha_{\mathrm{ub}})$; (iv) Condition 2; (v) Condition 3 at Huang's witness $\hat{z} = -0.6693$. Together with the strict decrease of $G_\star$, (i)–(iii) discharge Condition 1 verbatim on the slab: $\alpha_\star(\kappa)$ exists, is unique, and lies in the slab's α interval for every κ in the slab.

The contraction bound is claimed per slab only. Over a batch hull the κ width poisons the $\partial R/\partial q$ enclosure and the product of suprema loses a factor of about two against the supremum of the product; on a single slab the identity-based enclosure clears 1 with margin 0.049 at $\kappa = 0$, growing to 0.198 at $\kappa = -0.65$ and thinning to 0.0083 at the positive edge of the lane.

All 1500 slabs covering $[-0.65, 0.10]$ pass every check, in forty independent worker runs whose collector also gates contiguity (consecutive slabs must share their κ boundary). Two first attempts failed and are kept as evidence: one at doubled slab width (the q pad must cover the fixed point's drift across the slab's (α, κ) rectangle, dominated by $\partial q_\star/\partial\alpha \approx 0.92$ times the α pad; the direct drift $dq_\star/d\kappa$ stays near 0.18 across the strip), and one on the separated-form λ enclosure described above.

5.3 The Nakajima lane

The rectangle product bound saturates near $\kappa \approx 0.12$, and we measure that this wall belongs to the method, not the condition: at $\kappa = 0.13$ the box product reads 1.32, while a certified sign bisection of the displacement $P(R(q)) - q$ — which has certified signs at the interval endpoints and, by the uniqueness of the fixed point, a single zero between them, so it is positive below the fixed point and negative above — locates the fixed point at both α endpoints to width $2 \cdot 10^{-8}$ with no contraction input, the contraction evaluated on the located intervals reads 0.79, and the G crossing resolves with margins $2 \cdot 10^{-3}$.

The rectangle clause itself is then discharged at $\kappa = 0.13$ exactly as Huang's Proposition 3.2 discharges it at $\kappa = 0$: on the rectangle $\alpha \in [0.705932217, 0.705933898]$, $q \in [0.5836943, 0.5857104]$ (the located fixed points $\pm 10^{-3}$), the certified contraction product is 0.8435, the map is endpoint-inward at both α ends with margins above $2.3 \cdot 10^{-4}$, the G signs at the located fixed points are $+1.6 \cdot 10^{-7}$ and $-1.6 \cdot 10^{-6}$, and $\mathbb{E} \log \Psi < 0$ over the rectangle (`rect_cond31_0p13.json`; locate and G evaluator depth 16000). The wide-rectangle product 1.32 is a pad artifact, not a property of the map, and the map itself carries no wall across the lane: the true fixed-point contraction $(P \circ R_\alpha)'(q_\star)$ is nearly flat, rising only from 0.758 at $\kappa = 0.10$ to 0.770 at $\kappa = 0.19$, everywhere below 1 with margin above 0.22. What decides the rectangle clause is therefore the pad width, not the margin: a probe over the lane (`rect_strip_nak.jsonl`, per κ sub-ball) clears 1 at the lower edge with a $2 \cdot 10^{-3}$ half-pad but crosses it (1.012) by $\kappa = 0.15$ as that same pad inflates the flat 0.765, exactly as a thinner located rectangle pulls $\kappa = 0.13$ back to 0.844. The pad that matters is the located q interval, whose width is set mostly by the slab's α width $(3 \cdot 10^{-3})$ through $\partial q_\star/\partial\alpha \approx 0.9$; narrowing it means deep-locating $\alpha_\star$ per slab to $\sim 10^{-6}$, the same tightening the five point margins already carry, after which the flat contraction leaves a thin rectangle that clears 1. We ran this recipe at both ends of the lane: with $\alpha_\star$ deep-located to width $7 \cdot 10^{-6}$ and a $5 \cdot 10^{-4}$ half-pad, the rectangle clause certifies at $\kappa = 0.185$ (contraction 0.830, inwardness above $1.1 \cdot 10^{-4}$; `rect_cond31_0p185.json`) just as at $\kappa = 0.13$, so both lane endpoints carry the full Condition 1. Certifying every interior slab is those same two steps per slab — compute, not new mathematics, since the contraction the clause bounds never approaches its limit.

The second lane certifies $[0.10, 0.19]$ (180 slabs) on that design: existence by the same inward sweep; location by certified sign bisection per κ sub-ball (the κ width floors the displacement's sign resolution, so each slab runs eight sub-balls whose union covers it); the contraction consumed only on located intervals, which is the local form Huang's perturbation argument uses; and uniqueness from Nakajima's saddle theorem [5], whose premise $\alpha_{\mathrm{ub}} < \alpha_c(\kappa) = 2/(\pi\,\mathbb{E}[(\kappa - Z)_+^2])$ is certified per slab through the closed form $\mathbb{E}[(\kappa - Z)_+^2] = (1+\kappa^2)\big(1-\Psi(\kappa)\big) + \kappa\varphi(\kappa)$ (worst certified premise margin 0.311).

Across both lanes: 1680 contiguous slabs, zero failures. At five margins a certified bisection on the sign of G at located boxes tightens the slab intervals to width about $2 \cdot 10^{-6}$:

$$
\begin{array}{ll}
\alpha_\star(0.05) \in [0.781073068,\, 0.781074776], & \alpha_\star(-0.05) \in [0.889408032,\, 0.889409783], \\
\alpha_\star(0.0995) \in [0.733478514,\, 0.733480204], & \alpha_\star(-0.45) \in [1.557652103,\, 1.557654133], \\
\alpha_\star(0.13) \in [0.705932218,\, 0.705933898]. &
\end{array}
$$

Each interval contains its continuation value; the slab containing $\kappa = 0$ contains the Ding–Sun constant. The bisection floor is measured, not guessed: at the default quadrature depth the G enclosure binds at width $1.4 \cdot 10^{-5}$, and raising the two quadrature floors breaks through exactly as budgeted.

6 The center of the two-variable landscape

The local ingredient of Condition 4 is the negative definiteness of the Hessian M of Huang's first-moment functional at its degenerate maximizer. His second-derivative expansion, *before* the zero-margin specialization, gives at every κ the same quadratic form

$$
\langle u, Mu\rangle = -(I_1u_1^2 + 2I_2u_1u_2 + I_3u_2^2) + C_1a^2 + C_2ab + C_3b^2, \qquad a = I_1u_1 + I_2u_2,\; b = I_2u_1 + I_3u_2,
$$

with prior-side integrals I_j unchanged in form ($I_1 = \psi_0(1-q_0) - 2\psi_0^2(1 - 4q_0 + 3p_4)$, $I_2 = \psi_0(1-4q_0+3p_4)$, $I_3 = q_0 - p_4$, $p_4 = \mathbb{E}\tanh^4(\sqrt{\psi_0}Z)$) and constraint-side constants C_j that at $\kappa \neq 0$ must be taken from the pre-specialization display: his zero-margin closed forms rest on two identities of the unshifted law, of which $\alpha\,\mathbb{E}[N^2] = \psi_0$ survives at every margin (it is the same identity behind the unbounded-tilt directions of Section 7) while $\alpha\,\mathbb{E}[N\hat{H}] = -q_0(1-q_0)\psi_0$ does not. We evaluate the general-margin C_j directly at the shifted law. With $\hat{H} = \sqrt{q_0}\,Z$, $u = (\kappa - \hat{H})/\sqrt{1-q_0}$, $N = M(u)/\sqrt{1-q_0}$, $F' = -M'(u)/(1-q_0)$, and $Q = -\hat{H}/q_0 + \big(\kappa - \hat{H} - (1-q_0)N\big)/(1-q_0)$, every expectation over the single Gaussian $\hat{H}$,

$$
C_1 = \frac{\alpha}{\psi_0^2}\,\mathbb{E}[F'N^2], \qquad C_2 = -\frac{2\alpha}{\psi_0}\,\mathbb{E}[F'NQ] + \frac{2}{1-q_0},
$$

$$
C_3 = \alpha\,\mathbb{E}[F'Q^2] + \frac{2\alpha}{q_0(1-q_0)}\,\mathbb{E}[N\hat{H}] - \alpha\Big(\frac{3}{(1-q_0)^2} + \frac{1}{q_0(1-q_0)}\Big)\mathbb{E}\big[N\,(\kappa - \hat{H} - (1-q_0)N)\big].
$$

The certified evaluation encloses each expectation by the mean-value cell rule with explicit quadratic tail majorants (N grows at most linearly on the left tail, F' is bounded by $1/(1-q_0)$ in magnitude, and every integrand is degree at most two in $|z|$), and the near-cancelling prior moments are evaluated as single correlated integrands. The certified enclosures at the six margins are one-sided where sign is the claim:

$$
\begin{array}{lll}
\kappa = -0.45: & M_{11} \le -0.0647, & \det M \ge 2.9 \cdot 10^{-4}, \\
\kappa = -0.05: & M_{11} \le -0.0471, & \det M \ge 1.6 \cdot 10^{-4}, \\
\kappa = 0: & M_{11} \le -0.0451, & \det M \ge 1.8 \cdot 10^{-4}, \\
\kappa = 0.05: & M_{11} \le -0.0424, & \det M \ge 1.2 \cdot 10^{-4}, \\
\kappa = 0.0995: & M_{11} \le -0.0400, & \det M \ge 1.0 \cdot 10^{-4}, \\
\kappa = 0.13: & M_{11} \le -0.0385, & \det M \ge 8.8 \cdot 10^{-5},
\end{array}
$$

the trend — both margins thinning as κ grows — being the quantitative form of the landscape flattening that Section 9.1 prices.

Certification requires parameter boxes far tighter than the slabs provide ($\det M \sim 10^{-4}$ moves at order $dC/dq \sim 10$), which is what the deep $\alpha_\star$ intervals above are for. On the resulting located boxes ($3.6 \cdot 10^{-6}$ in q_0, $5.2 \cdot 10^{-5}$ in ψ_0 at $\kappa = 0.05$) the certificate resolves: $M_{11} < 0$ and $\det M > 0$ in ball arithmetic, with the near-cancellations $1-4q_0+3p_4$ and $q_0 - p_4$ evaluated as single correlated integrands and the five constraint-side integrals by mean-value Riemann sums with elementary Mills-family derivatives. The same chain certifies at $\kappa = 0$ on the Ding–Sun rectangle — where the C_j enclosures land inside Huang's certified intervals, an independent re-verification of his claim through the general-margin formulas — and at -0.05, 0.0995, -0.45 (where $C_1 = -0.855$ against -0.718 at zero margin: genuinely shifted territory, not a perturbation), and 0.13 past the product-bound wall. Six margins in all.

Along the continuation the fixed-tilt M stays negative definite up to $\kappa = 1.15$ and loses definiteness inside $(1.1812, 1.1844)$, while the Hessian of the tilt-optimized functional remains negative out to at least $\kappa = 1.5$; the two are consistent by the Schur relation (the fixed-tilt form dominates), so past $\kappa \approx 1.18$ a local argument must re-optimize the tilt, as the ray certificates below already do.

7 The two-variable function at general margin

Huang's rate function at margin κ, in the form his proof consumes, is

$$\mathcal{S}(\Lambda, s) = \frac{s^2\psi_0}{2} + \mathrm{ent}(\Lambda) + \alpha\,\mathbb{E}\log\Psi\left(\frac{\kappa - (a_2/q_0)\hat{H} - (a_1/\psi_0)N}{D} + sN\right),$$

where $\dot{H} \sim N(0,\psi_0)$ and $\hat{H} \sim N(0,q_0)$ are independent, $M = \tanh\dot{H}$ (from here on M is this variable, following Huang; the ratio φ/Ψ of the earlier sections appears from now on as the hazard function E), $N = F_{1-q_0}(\hat{H}) = E\big((\kappa-\hat{H})/\sqrt{1-q_0}\big)/\sqrt{1-q_0}$ is the shifted cavity law (this is one of the two places the margin enters; the other is the additive κ in the $\log\Psi$ argument), $a_1 = \mathbb{E}[\dot{H}\Lambda]$, $a_2 = \mathbb{E}[M\Lambda]$, $D = \sqrt{1 - a_2^2/q_0}$, and $\mathcal{S}_\star(\lambda_1,\lambda_2) = \inf_{s\ge 0}\mathcal{S}(\tanh(\lambda_1\dot{H} + \lambda_2 M), s)$. Condition 4 asks $\mathcal{S}_\star \le 0$ everywhere.

The zero-margin verification's reduction applies verbatim: the profile enters only through the moment pair (a_1, a_2), which ranges over a compact convex body K (the moment body of $(\dot{H}, M)$), and on K the entropy term is bounded by convex duality, $H(a) \le \Phi(b) - b\cdot a$ for every dual point b, with $\Phi(b) = \mathbb{E}\log 2\cosh(b_1\dot{H} + b_2 M)$ free of κ. The margin lives entirely in the constraint term. Two structural facts organize the verification. The maximizer sits at $a^\star = (\psi_0(1-q_0), q_0)$ for every κ: this is the base-point identity $\nabla\Phi(1,0) = (\mathbb{E}[\dot{H}M], \mathbb{E}[M^2]) = (\psi_0(1-q_0), q_0)$ (Gaussian integration by parts and the fixed-point equations), and Huang's lemma gives $\mathcal{S}_\star = 0$ there with optimal tilt $s_0 = \sqrt{1-q_0}$, both in closed form. And on roughly the half-plane $\lambda_1 < 0$ the s-infimum is genuinely $-\infty$: $\alpha\,\mathbb{E}[N^2] = \psi_0$ exactly at the fixed point, so the quadratic tilt coefficient cancels and a subleading linear term runs away; those directions satisfy the condition trivially and are the capped-tilt case in Huang's treatment.

Float diagnostics along the curve (the certified analogue follows) show a landscape essentially independent of κ: a single peak at $(1, 0)$, off-center values below $-3\cdot 10^{-3}$, negative far rays, Hessian eigenvalues near $(-0.22, -0.0045)$, and moment-body clearance near 0.017, with the weak eigenvalue degrading toward deep negative margin (-0.0013 at $\kappa = -1.5$) — the measured approach to the expected replica-symmetry-breaking territory.

8 Certifying the two-variable condition at $\kappa = 0.05$

The zero-margin certificate has four pieces: a sweep of the moment body outside an exclusion box around $a^\star$ (Region II), banded ray-concavity certificates on a star around $a^\star$ (Region I), a second sweep of the exclusion box minus the star (stage 2), and an interior-ball certificate placing the star strictly inside the moment body. The obstruction at $\kappa \neq 0$ is uniform across all four: at zero margin the parameters (q_0, ψ_0) are certified to a rectangle of width 10^{-9}, so the maximizer's coordinates can be stored as decimals and every enclosure treats them as exact; at $\kappa = 0.05$ the located boxes have width up to $1.9 \cdot 10^{-4}$ in ψ_0, and that width enters every plain evaluation of Φ at about $2 \cdot 10^{-5}$ — fatally wide next to valley margins of order 10^{-7} in the dual localization. Three devices repair the architecture; each is exact mathematics, not tuning.

The symbolic maximizer. Every certificate that references $a^\star$ or the base tilt s_0 consumes them as the parameter balls $(\,\psi_0(1-q_0),\, q_0\,)$ and $\sqrt{1-q_0}$ — never as stored decimals — so the true maximizer, at the true parameters, is covered exactly and the ray identities ($\mathcal{S} = 0$, $\nabla\mathcal{S} = 0$, tilt optimality at the true $a^\star$) hold on the certified rays with no origin-inflation ball. Ray coverage is exact: certificates evaluated over the origin ball cover in particular the rays from the true origin.

The correlated Bregman evaluator. The dual localization (the argument that the dual maximizer $\lambda(x)$ stays inside a certified box, so that the entropy Hessian can be bounded there) needs signs of Bregman gaps $\Phi(\lambda) - \Phi(\hat\lambda_k) - (\lambda - \hat\lambda_k) \cdot x$ with $x = a^\star + \text{offset}$. Evaluated as differences of plain Φ enclosures these carry the full parameter fuzz twice. Written against the anchor $b_0 = (1, 0)$, the integrand collapses exactly: with $\Delta = (\lambda_1 - 1)\dot H + \lambda_2 M$,

$$\log \frac{\cosh(\dot H + \Delta)}{\cosh \dot H} - (\lambda - b_0) \cdot \nabla_b \log 2\cosh(b \cdot (\dot H, M))\big|_{b_0} = \log\bigl(\cosh\Delta + M \sinh\Delta\bigr) - M\Delta,$$

because $(\lambda - b_0) \cdot \nabla h(b_0) = M\Delta$ pointwise. Every factor is a small ball with thin coefficients, so the parameter width enters only at second order in $|\lambda - b_0|$; at the box corners that used to fail, the measured enclosures tighten by two orders of magnitude, and at the anchor the evaluator returns the exact zero ball. Bregman gaps between arbitrary points are then differences of two anchored gaps, and the gradient term the localization needs is the same trick applied to $\nabla\Phi$: $\nabla\Phi(\lambda) - a^\star$ has the exact per-node form $(\dot H, M)\,(1 - M^2)\tanh\Delta/(1 + M\tanh\Delta)$.

The factor $\cosh\Delta + M\sinh\Delta$ itself has two exact evaluations with disjoint stability regimes: the cosh/sinh form is exact at $\Delta = 0$ (which the inner-band gaps of order 10^{-7} require) but subtracts two $e^{|\Delta|}$-sized balls when $M \approx \pm 1$ and $|\Delta|$ is large, where the guard against the complex logarithm's branch then fires on every subdivision; the split $e^{\Delta}(1+M)/2 + e^{-\Delta}(1-M)/2$ has both terms nonnegative on the real axis but carries first-order M-width near $\Delta = 0$. The evaluator branches per node at $|\Delta| = 1$; both forms are identities, so the branch affects tightness only.

Stable far-tail kernels. The ray certificates consume second derivatives of the constraint term, whose kernels involve $E'(V) = E(V)(E(V) - V)$ and $E''(V)$ for the hazard function $E = \varphi/\Psi$ evaluated at cavity arguments V that reach $O(20)$ on far quadrature cells; there $E(V) - V$ is a difference of two large balls and the kernels explode (we measured certified upper bounds of $+2 \cdot 10^3$ on cells whose true ray second derivative is -6.6). Three classical facts, applied as intersections with the computed balls, remove the failure mode entirely:

$$0 < E(V) - V < \frac{1}{V} \ \ (V > 0), \qquad E' \in (0,1), \qquad 0 < E'' < E - V,$$

the second because the one-sided truncated Gaussian has variance $1 - E'(V) \in (0,1)$, the third following from convexity of the Gaussian hazard together with $E'' = E'\left(E + (E - V)\right) - E < E - V$ when $E' < 1$. Intersecting a computed enclosure with a proven bound is always sound; on benign cells the computed ball is returned unchanged, and the $\kappa = 0$ regression is digit-identical.

With these three devices the four pieces certify at $\kappa = 0.05$, on the located boxes and the deep $\alpha_\star$ interval of Section 5:

(i) *Region II*: 1296 top cells over the moment rectangle minus the exclusion box, refining to 867 leaves, zero failures.

(ii) *Region I*: the banded ray-concavity certificates over the star (radius 0.012 along the weak direction), 1404 band jobs from the symbolic origin, zero failures; the thinnest certified band margin is $-4.7 \cdot 10^{-5}$ and the deepest $-1.6 \cdot 10^2$. A supplementary run of twelve shoulder chunks extends the certified reach to 0.009 on the arcs 0.14 to 0.19 radians off the weak axes (margins -0.35 to -2.7): the zone policy, not the concavity, was the binding constraint there, and stage-2 leaves straddling the wedge shoulder need the extra reach.

(iii) *Stage 2*: the exclusion box minus the star, 1365 leaves, zero failures, with star containment read from the Region-I manifests themselves — a certified band covers every direction of its chunk out to the star boundary, so the certified reach is the pointwise union of the main zone reach and the supplementary arcs, evaluated exactly on a piecewise-constant splitting; the SHA-256 of both manifests is bound into the stage-2 artifact.

(iv) *Star interior*: the star, origin ball included, lies strictly inside the moment body: constructive inradius 0.01481 against required 0.01209.

Together: $\mathcal{S}_\star \le 0$ on the whole plane at $\kappa = 0.05$. All four of Huang's hypotheses hold there, and his theorem gives the unconditional bound.

Theorem 1. *The storage capacity of the Ising perceptron at margin κ is at most $\alpha_\star(\kappa)$ for every $\kappa \in \{-0.45, -0.05, 0.05, 0.0995, 0.13\}$, with the certified enclosures*

$$\begin{aligned} &\alpha_\star(-0.45) \in [1.557652103, 1.557654133], && \alpha_\star(-0.05) \in [0.889408031, 0.889409783],\\ &\alpha_\star(0.05) \in [0.781073068, 0.781074776], && \alpha_\star(0.0995) \in [0.733478513, 0.733480205],\\ &\alpha_\star(0.13) \in [0.705932217, 0.705933898] : \end{aligned}$$

for α above the corresponding interval, with probability $1 - o(1)$ no $w \in \{-1, +1\}^N$ satisfies all αN constraints.

The same pipeline is parameterized over the located boxes of any certified slab; what stands between the point theorem and a κ-interval version of Condition 4 is quantified in Section 9.1. The other three conditions do not wait for that extension: they are certified over the whole strip, which gives the conditional structure of the program a closed form.

Theorem 2. *For every $\kappa \in [-0.65, 0.10]$, Conditions 1–3 hold, with the uniform rectangle contraction of Condition 1 certified on the slab rectangle itself. For every $\kappa \in (0.10, 0.19]$, the existence, uniqueness, and crossing of Condition 1 hold per slab (uniqueness from Nakajima's theorem under a certified premise), and Conditions 2 and 3 hold per slab; the rectangle contraction is additionally certified, on a deep-located thin rectangle, at the lane endpoints $\kappa \in \{0.13, 0.185\}$. In every case $\alpha_\star(\kappa)$ exists, is unique on the fixed-point branch, and lies in the certified α interval of the slab containing κ. Consequently, at every κ at which Condition 4 and the full Condition 1 both hold — in particular the five margins of Theorem 1 — and for every α above that slab interval's upper endpoint, with probability $1 - o(1)$ no $w \in \{-1, +1\}^N$ satisfies all αN constraints at margin κ.*

Condition 4 is certified at the five margins of Theorem 1 and at $\kappa = 0$, where our pipeline independently reproduces the zero-margin verification; at those six values the hypothesis of Theorem 2 discharges and the two theorems agree. Everywhere else on the strip, the entire distance between the conditional and unconditional statements is one two-variable inequality at that single margin.

9 Further margins

The four conditions of Theorem 1 are margin-wise statements, and the certificate pipeline of Sections 7–8 runs unchanged at any κ inside the strip where the fixed point is certified unique: the pilot locates the (q_0, ψ_0) boxes at both endpoints of the α window, configures the exact-arithmetic modules from those balls, and runs the same four pieces. Beyond $\kappa = 0.05$ we have certified Condition 2varfn at $\kappa = -0.05$, where the fixed-point uniqueness input comes entirely from the contraction certificates of the companion strip (there is no independent derivation on the negative side). The certified pieces at both margins:

κ	$\alpha^*(\kappa)$ window	sweep	Region I	stage 2	inradius / required
0.05	[0.781073068, 0.781074776]	867	1404 (+12)	1365	0.01481 / 0.01209
−0.45	[1.557652103, 1.557654133]	655	1404 (+12)	1196	0.01480 / 0.01207
−0.05	[0.889408031, 0.889409783]	884	1404	1294	0.01527 / 0.01209
0.0995	[0.733478513, 0.733480205]	904	1404 (+12)	1394	0.01459 / 0.01209
0.13	[0.705932217, 0.705933898]	883	1404 (+12)	1421	0.01446 / 0.01210

Table 1: Certified leaf and band counts per piece, with zero failures in every manifest. The 0.05 Region I count includes the twelve shoulder-supplement chunks; at −0.05 the stage-2 lattice clears the wedge shoulder without a supplement. The final column gives the constructive moment-body inradius against the required star radius, both exact-arithmetic bounds.

At $\kappa = -0.05$ the manifest also carries per-band interval packets for the frozen-tilt matrix B, and the kernel-checked skeleton gains a fifth family: Loewner definiteness of every band's B-matrix, decided over the integers in thirteen chunks of 108 bands together with a decided total count (55 kernel theorems at this margin; the mutation battery rejects a sign corruption in each family).

The 0.13 run additionally exposed a completeness defect in the stage-2 containment—supplement arcs stored on one angular branch were invisible to cells whose interval fell on the other, weakening containment only, never soundness—which was fixed and the margin re-certified with zero failures. One observation from the 0.0995 run is also methodologically relevant. Its Region II sweep initially failed a single minimal-size cell just above the exclusion box's upper edge, where the dual certificate degenerates in a thin ridge (the certified value interval collapses to $[\pm 0.181]$ on a cell of width 0.0018, recovering two cell-widths higher). The exclusion box's upper offset moved from $+0.0961$ to $+0.1030$ in response; this is sound without revisiting Region I because the stage-2 containment derives its tiling from the same box and fails loudly if the enlarged region outgrows what the star and supplement cover. The 0.05 stage-2 was re-run against its published Region I manifest under the enlarged box on two machines independently; both certify with 1365 leaves and zero failures, and the shipped manifest is the box run with its bindings intact.

9.1 From points to intervals

The natural strengthening of Theorem 1 replaces each margin by a κ-interval, so that $\alpha_\star(\kappa)$ becomes an unconditional bound on a continuum. For Conditions 1–3 this is what the strip certificates already provide: those inequalities hold with margins of order 10^{-1}, and enclosing every input over a κ-slab costs far less than that. Condition 4 is different in kind, and it is worth recording precisely why, because the obstruction is quantitative and, we believe, intrinsic.

Certifying $S_\star \le 0$ over a slab means every certificate must hold with the located parameters (α, q_0, ψ_0) replaced by their hulls across the slab. Three measurements fix the scales. First, the hull widths cannot be made small: the located enclosures carry irreducible floors (the inward fixed-point test resolves no finer than the quadrature fuzz, about $3.5 \cdot 10^{-4}$ in q_0, and $\alpha_\star$ itself drifts at $|d\alpha_\star/d\kappa| \approx 1$–$3.4$ across any slab), which translate into a moment-plane origin ball of radius at least a few times 10^{-3} however narrow the slab. Second, the evaluation fuzz of every certificate grows linearly with that ball: rerunning the $\kappa = -0.65$ sweep with slab-hulled parameters (ball $2.3 \cdot 10^{-2}$) inflates cell enclosures to ± 1.3, and the scaling back to the floor ball leaves ± 0.2. Third, the certificates that matter are thin: the sweep's weakest certified cells at point margins sit at $-5 \cdot 10^{-4}$, and Region I's thinnest bands at $-4.7 \cdot 10^{-5}$, because the two-variable function is genuinely flat at its degenerate maximizer. A 10^{-4}-margin certificate cannot survive 10^{-1}-scale fuzz, at any slab width, with this evaluator architecture.

The same comparison explains why the strip conditions posed no such problem and points at the correct construction. Near the maximizer the certified Hessian gives quadratic control whose constants vary slowly in κ (the fixed-tilt determinant moves by about $2 \cdot 10^{-4}$ per unit margin), so a κ-uniform neighborhood estimate is available from point certificates plus a crude κ-derivative bound — crude is enough, because a Lipschitz constant needs no thin margin. Far from the maximizer the certificates are fat and absorb hull fuzz directly. What remains is the intermediate annulus, where quadratic control must be extended beyond the star radius with explicit remainder bounds before the two regimes meet. We record this as the open engineering of the margin program: the obstruction is measured, the two easy regimes are in hand, and the annulus estimate is the one genuinely new ingredient an interval theorem requires.

The certified artifacts also price that ingredient. Across the three positive margins the Region I band populations have nearly identical margin profiles: at most one or two bands per margin certify below $3 \cdot 10^{-4}$, at most seven below $3 \cdot 10^{-3}$, and more than 95 percent above $3 \cdot 10^{-2}$. If each band carried a certified bound L_i on the κ-derivative of its certificate value, a band would hold over a κ-radius m_i/L_i of its margin m_i, and re-certifying each band at its own pitch would cover 10^{-2} of margin axis for 1.1 times the cost of one point run at unit derivative bound, 1.3 times at a conservative bound of three — against roughly $3 \cdot 10^5$ band evaluations for uniform slabs at the thinnest band's pitch. The compute is therefore not the obstruction; what is missing is the derivative enclosure itself, one certified κ-derivative per certificate family, with the located branch differentiated implicitly. The constant itself is measurable from the existing artifacts: the band lattice is shared across margins up to the wedge rotation, and 1292 rectangles coincide exactly between the $\kappa = 0.05$ and $\kappa = 0.0995$ manifests, giving certified finite differences of each certificate value along the located branch. The measured sensitivities are strongly in the plan's favor: the median band moves at $|\Delta S/\Delta\kappa| \approx 6$, but the thin-trough bands move at 0.1–0.3, so the thinnest band (margin $5.6 \cdot 10^{-4}$) already holds over a κ-radius near $2 \cdot 10^{-3}$, and with per-band measured pitches the count is 1.04 runs per 10^{-2} of margin axis — about eight Region I run-equivalents for all of $[0.05, 0.13]$. A difference over a 0.05 gap bounds the mean drift rather than the local derivative, so these numbers size the project rather than prove it; the machinery an interval theorem still requires is the certified per-certificate κ-derivative, with the same measurement then repeated for the sweep, union, and star-interior families.

10 Toward the matching lower bound

Krauth and Mézard conjectured the value of the limiting ratio $M_N(\kappa)/N$ for every margin, not only at $\kappa = 0$ [1]; the constant 0.833 is the zero-margin point of the conjectured curve $\alpha_\star(\kappa)$. At $\kappa = 0$ the conjecture is resolved: Ding and Sun [3] proved $\alpha_\star$ is a lower bound with positive probability, the sharp-threshold sequence of Nakajima and Sun [11] upgrades positive probability to convergence in probability, Huang [4] proved the matching upper bound, and the numerical hypotheses of both proofs are certified [6]. Away from $\kappa = 0$ the upper half is Theorem 1; the lower half has, at present, no analytic theorem to certify. This section delimits the gap: a lemma on why unconditioned moment methods cannot close it at nonnegative margins, an embedded bound that does give nonzero-margin capacity lower bounds on the negative side, and a numerical verification that the κ-dependent inputs of the Ding–Sun program hold at all five certified margins.

10.1 Plain second moments fail at every margin

Write $Z = Z_{N,\kappa}$ for the number of $w \in \{-1, +1\}^N$ satisfying all $M = \lceil \alpha N \rceil$ constraints, and $q_\rho(\kappa) = \mathbb{P}(X \geq \kappa,\, Y \geq \kappa)$ for (X, Y) bivariate standard Gaussian with correlation ρ.

Lemma 3. *For every $\kappa \in \mathbb{R}$ and $\alpha > 0$,*

$$\liminf_{N\to\infty} \frac{1}{N} \log \frac{\mathbb{E}[Z^2]}{(\mathbb{E}Z)^2} \;\geq\; c(\alpha, \kappa) \;>\; 0.$$

Consequently the Paley–Zygmund bound applied to Z gives only $\mathbb{P}(Z > 0) \geq e^{-cN+o(N)}$, and no bound depending on $\mathbb{E}Z$ and $\mathbb{E}[Z^2]$ alone can do better, since a distribution placing mass $(\mathbb{E}Z)^2/\mathbb{E}[Z^2]$ at $\mathbb{E}[Z^2]/\mathbb{E}Z$ and the rest at zero matches both moments. For $\kappa \geq 0$ the same conclusion holds for every weighted count $Z_f = \sum_w \prod_a f(\langle g_a, w\rangle/\sqrt{N})$ with $f \geq 0$ supported on $[\kappa, \infty)$ and $0 < \int f \, d\Phi$.

Proof. $\mathbb{E}Z = 2^N \Psi(\kappa)^M$ by independence over constraints. For a pair w, w' at overlap ρ the constraint fields are bivariate Gaussian with correlation ρ, so counting ordered pairs by Hamming distance k,

$$\mathbb{E}[Z^2] = 2^N \sum_{k=0}^{N} \binom{N}{k} q_{1-2k/N}(\kappa)^M.$$

Fix $\rho_0 \in (0,1)$, let $k^* = \lceil N(1-\rho_0)/2 \rceil$ and $\rho^* = 1 - 2k^*/N$. Keeping the single term k^* and using $\binom{N}{k} \geq e^{NH(k/N)}/(N+1)$ with H the natural-log binary entropy ($H(p) = H(1-p)$, so the exponent may be written with $(1+\rho_0)/2$),

$$\frac{1}{N} \log \frac{\mathbb{E}[Z^2]}{(\mathbb{E}Z)^2} \geq \underbrace{H\Big(\tfrac{1+\rho_0}{2}\Big) - \log 2 + \alpha\big[\log q_{\rho_0}(\kappa) - 2\log \Psi(\kappa)\big]}_{=:\, \psi_{\alpha,\kappa}(\rho_0)} + O(1/N),$$

the error collecting the binomial prefactor, the $|\rho^* - \rho_0| \leq 2/N$ continuity of $\rho \mapsto \log q_\rho$ (differentiable, below), and $|M - \alpha N| \leq 1$. Now $q_0 = \Psi(\kappa)^2$ and $H(1/2) = \log 2$ give $\psi_{\alpha,\kappa}(0) = 0$. Reflecting the orthant to a joint CDF, $q_\rho(\kappa) = \Phi_2(-\kappa, -\kappa; \rho)$, and Plackett's identity $\partial\Phi_2/\partial\rho = \varphi_2$ with the symmetry $\varphi_2(-\kappa,-\kappa;\rho) = \varphi_2(\kappa,\kappa;\rho)$ give $\partial q_\rho/\partial\rho = \varphi_2(\kappa,\kappa;\rho)$ on $(-1,1)$; at $\rho = 0$ this is $\varphi(\kappa)^2$. Since $\frac{d}{d\rho} H(\frac{1+\rho}{2}) = \frac{1}{2}\log\frac{1-\rho}{1+\rho}$ vanishes at 0,

$$\psi'_{\alpha,\kappa}(0) = \alpha\, \frac{\varphi(\kappa)^2}{\Psi(\kappa)^2} \;>\; 0,$$

so $\psi_{\alpha,\kappa}(\rho_0) \geq \rho_0\, \psi'(0)/2 > 0$ for all small $\rho_0 > 0$; fix such a ρ_0 and set $c = \psi_{\alpha,\kappa}(\rho_0)$. For the weighted counts, Mehler's expansion gives $\mathbb{E}[f(X)f(Y)] = \sum_{j\geq 0} \rho^j \langle f, \mathrm{He}_j\rangle^2/j!$, whose ρ-derivative at 0 is $(\int x f \varphi\, dx)^2$; for $f \geq 0$ supported on $[\kappa, \infty)$ with $\kappa \geq 0$ the integrand $xf(x)\varphi(x)$ is nonnegative, and it vanishes almost everywhere only if $f\, d\Phi$ concentrates at $\{0\}$, which $0 < \int f\, d\Phi$ together with $\mathbb{E}Z_f > 0$ excludes; the argument then runs unchanged. □

The mechanism is the asymmetry: pairs of solutions at small positive overlap are exponentially favored over independent pairs at every margin, which is what the AMP-conditioned second moment of [3] is built to circumvent at $\kappa = 0$, and what any lower-bound proof must circumvent at every other margin. The zero-margin instance of this failure is classical (see the introduction of [3]); the content here is the uniform mechanism and the closure over nonnegative-margin weights. For $\kappa < 0$ the weighted closure genuinely fails, and profitably so: a window weight symmetric about the origin has zero linear Mehler coefficient, which is the next observation.

10.2 A capacity lower bound at negative margins

Proposition 1. *Let $\kappa < 0$ and let $\alpha_c^{\mathrm{sym}}(K) = \log 2/(-\log \mathbb{P}(|X| \leq K))$ denote the symmetric binary perceptron threshold at width K. For every $\alpha < \alpha_c^{\mathrm{sym}}(|\kappa|)$, with probability $1 - o(1)$ the margin-κ perceptron admits a solution: $M_N(\kappa)/N \geq \alpha$ whp.*

Proof. If $|\langle g_a, w\rangle|/\sqrt{N} \leq -\kappa$ for all a then in particular $\langle g_a, w\rangle/\sqrt{N} \geq \kappa$, so the symmetric solution set at width $|\kappa|$ embeds in the margin-κ solution set. The symmetric threshold is α_c^{sym} [7], with the satisfiability side holding with high probability by the lognormal limit of Abbe, Li, and Sly [9] (see also Perkins and Xu [8]). □

The bound is far from sharp — at $\kappa = -0.05$ it gives 0.215 against the predicted 0.889, at $\kappa = -0.45$ it gives 0.655 against 1.558 — but it is, as far as we know, the only rigorous nonzero-margin lower bound presently available for the one-sided Ising perceptron, and it is strictly positive on the whole negative axis.

10.3 The Ding–Sun condition at general margin

The construction underlying the $\kappa = 0$ lower bound of [3] is written at general margin throughout their Section 2: their equations (34)–(48) define, for parameters $(\kappa_{\text{all}}, \alpha_{\text{all}})$, an explicit univariate function $S_{\kappa,\alpha}(\lambda)$ on $[\lambda_{\min}, 1]$ built from the fixed point $(q_\star, \psi_\star)$ — an entropy term through the two-spin measure $P_{H,D}$, a constraint term through the conditional Mills law φ_ξ, and a convex $\inf_s$ correction — and their Condition 1.2 is the statement $S_{0,\alpha_\star}(\lambda) < 0$ for $\lambda \notin \{0, 1\}$. The κ-analog of that hypothesis is therefore already defined in their own notation, and every κ-dependent ingredient it consumes is an object this paper certifies: the fixed point, its uniqueness rectangle, and the crossing (Theorem 2 and the deep enclosures of Theorem 1).

We evaluated $S_{\kappa,\alpha_\star(\kappa)}$ in floating point at the five certified margins and at $\kappa = 0$ (`ds_condition.py`; Gauss–Hermite and conditional Gauss–Legendre quadrature, with the seven endpoint identities $H(0) = P(0) = A(0) = 0$, $H(1) = -H_\star$, $P(1) = -P_\star$, $G_\star = H_\star + P_\star \approx 0$, with $H_\star, P_\star$ the entropy and constraint halves of the free energy in their notation, and the conditional-mean check resolving at 10^{-12} or better at every margin as implementation controls). The result is uniform (Figure 2): at every margin the curve is strictly negative off $\{0, 1\}$ with its interior maximum at most $-5 \cdot 10^{-5}$, the maximum at $\lambda = 0$ is degenerate exactly as at $\kappa = 0$, and the whole family collapses nearly onto a single shape, with $\lambda_{\min}$ drifting from -0.441 at $\kappa = -0.45$ to -0.421 at $\kappa = 0.13$. These are diagnostics, not certificates; certifying $S_{\kappa,\alpha_\star} < 0$ off the roots is a univariate problem of exactly the species the Condition 4 machinery already handles, and we expect it to be far easier than the two-variable certification carried out here.

What separates these verified inputs from a general-margin lower bound is the probabilistic body of [3]: the AMP-conditioned moment computation of their Sections 2–7, whose displays carry κ_{all} explicitly but whose proofs are executed at $\kappa = 0$. An audit of that body would need, beyond the certified inputs above: a priori bounds at margin (the upper one is free at every κ from the first moment, since $\mathbb{E}Z \to 0$ exponentially above the annealed threshold $-\log 2/\log \Psi(\kappa)$; the lower one is Proposition 1 for $\kappa < 0$ and open for $\kappa > 0$), the state-evolution inputs (generic in the nonlinearity, hence in κ), and the sharp-threshold transfer of [11] if convergence in probability rather than positive probability is wanted. We record this as the concrete route to the exact threshold at the certified margins: the numerical side is done at six points and the analytic side is a κ-generalization audit of an existing proof, not a new construction.

11 Verifying the verification

A verification of this size is itself software, and we treat its correctness as part of the result. Five independent layers check the chain, and on the zero-margin verification these layers caught disjoint defect classes, which is the design intent.

Ball-arithmetic soundness by construction. Every quantity is an arb ball containing the true value; a PASS is a ball comparison true only when certain. The primitives (monotone

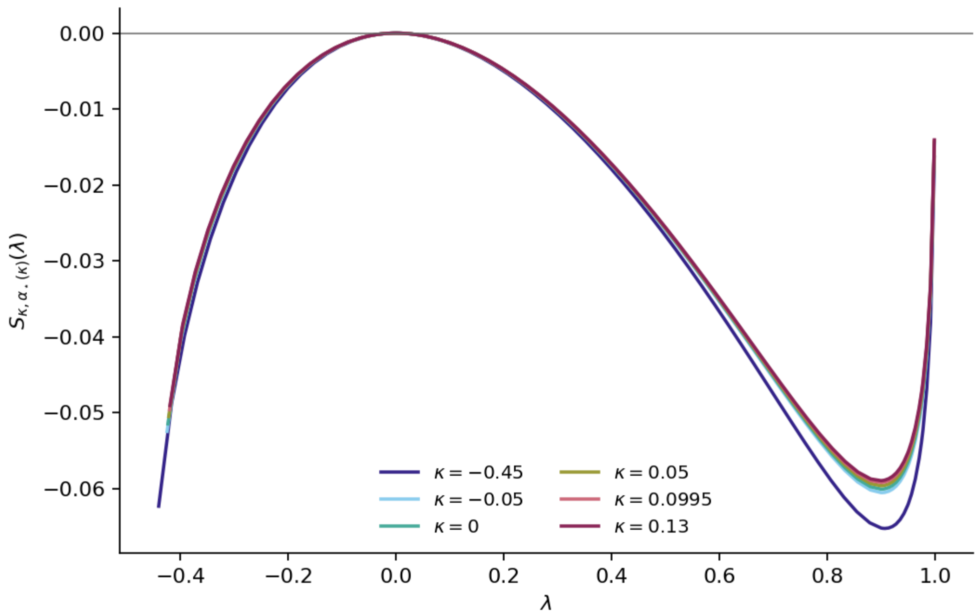

Figure 2: The general-margin Ding–Sun lower-bound condition, evaluated at the five certified margins and $\kappa = 0$: the univariate function $S_{\kappa,\alpha_\star(\kappa)}$ of their equation (48), whose nonpositivity off $\{0, 1\}$ is the numerical hypothesis of their second-moment argument. All six curves are negative off the two roots and nearly coincide.

endpoint evaluations, mean-value cells, correlated integrands, tail bounds) are shared with the audited zero-margin verification.

Symbolic verification of every identity. The stationarity and $\partial R/\partial q$ identities, the λ reformulation (identity, monotonicity, bound), the Nakajima closed form, the general-margin additions to the constraint kernels, and the Bregman collapse are checked in computer algebra; the finite-difference anchors run alongside.

Anchors at $\kappa = 0$. Every pipeline reproduces the certified zero-margin values: the fixed-point rectangle, Huang's AMP and λ constants, his C_j and $\det M$ intervals, and the ported quadrature layer is digit-identical against the zero-margin original at test points.

Exact artifacts and replay. Certificates carry every sign-critical quantity as outward-rounded decimal packets in canonical JSON, with SHA-256 source and payload binding and atomic no-clobber publication; the stage-2 artifact binds the hash of the Region-I manifest it consumed. (Some schedule-geometry fields remain ordinary JSON floats; they parameterize display and tiling bookkeeping, not the inequalities.) A documentation self-check keeps the replay instructions honest, and the fresh-clone consumer test (clone cold, run as a reproducer would) is part of the release procedure — on the zero-margin release it caught a stale generator that no other layer could see. A dependency-free portable checker re-verifies the connection layer from the JSON artifacts alone: the envelope hashes, the stage-2-to-Region-I bindings, the star-interior chain recomputed outward in exact rationals, and the tilings within each serialized chain. The checker also enforces two global completeness clauses on the band geometry: the innermost ring must close the full circle as a single seam-exact run, and each level's angular support must sit inside the previous level's, which together make every direction's radial chain contiguous out to the radius where its support ends — a deliberately corrupted manifest with all hashes consistently rebound fails exactly this check. Binding the outermost radius to the consuming stage's stated requirement awaits a schema field the current manifests predate, and the Lean tilings carry the same per-chain scope; both are recorded as open hardening.

A second proof system. The exact combinatorial and algebraic layer of the certificate chain — the star-interior inequality chain and the positivity of its perturbation determinant from the serialized packets, the definiteness of the Loewner majorants, and the gap-free tilings of the sweep grids and the Region-I band schedule — is re-stated over the integers and kernel-checked in Lean 4, following the zero-margin release: per-margin generated files whose kernel-decided theorems cover the 1404-band radius chain and its angular runs, both sweep grid tilings, and the star-interior packet chain, with the Loewner-majorant family at the four margins whose manifests carry exact B-matrix packets (the $\kappa = 0.05$ manifest predates that schema and its regeneration is queued). The extraction is mirrored in exact rational arithmetic so an extraction bug cannot silently weaken the Lean claims, and a mutation battery (negating one certificate integer per file) is rejected four for four. The slab-level sign claims join this layer as their manifests are regenerated with exact packets (the emitters now serialize them; the original strip manifests predate that and carry display strings only). This layer deliberately does not formalize the analytic mathematics; it removes the Python layer from the trusted base for the parts a kernel can decide.

12 What the curve says about learning

For binary weights the Euclidean norm is fixed at $\sqrt{N}$, so a margin constraint is exactly a robustness certificate: if $y_a\langle w, x_a\rangle/\sqrt{N} \geq \kappa$ then the label y_a is invariant under every perturbation of x_a of Euclidean norm below κ. With centrally symmetric i.i.d. designs and independent labels, label folding makes the robust-interpolation event identical to the satisfiability event studied here, so $\alpha_\star(\kappa)$ is the phase boundary for how many random labels per parameter a one-bit-per-parameter linear classifier can memorize with prescribed robustness radius. The certified $\kappa = 0.05$ bound, for instance, says such a classifier cannot robustly store more than 0.78108 random labels per parameter at that radius, with the guarantee holding against a worst-case adversary in feature space.

Reading the whole curve quantitatively (`results/robustness_price.csv`): on the certified strip the threshold decays log-linearly at rate 1.39 per unit margin — robust storage capacity halves for every 0.50 of demanded feature-space radius — and the decay continues smoothly beyond (a factor 4.8 from $\kappa = 0$ to 1.5). The efficiency of binary relative to real-valued weights *rises* with the demanded robustness: $\alpha_\star/\alpha_{\text{spherical}}$ grows from 0.417 at $\kappa = 0$ to 0.556 at $\kappa = 1.5$, so one-bit parameters give up proportionally less storage precisely where the robustness requirement is stiffest. On the relaxed side ($\kappa < 0$, tolerating margin violations up to $|\kappa|$) capacity grows toward the annealed bound (the ratio reaches 0.996 by $\kappa = -2$) and the landscape measurably approaches the replica-symmetry-breaking regime (the weak Hessian eigenvalue of the rate function degrades threefold by $\kappa = -1.5$), which is where the replica-symmetric prediction itself should eventually fail. The replica-symmetric cavity picture also predicts the margin distribution *among* solutions: conditionally on the quenched component $\sqrt{q}\,Z$ of a constraint's field, the field on a typical solution is $N(\sqrt{q}\,Z,\, 1-q)$ truncated to $[\kappa, \infty)$, so the gap density across constraints is the mixture $p(h) = \mathbb{E}_Z[\varphi_{1-q}(h - \sqrt{q}\,Z)\,\mathbf{1}\{h \geq \kappa\}/\Psi(u(Z))]$.

Computed along the branch at $\kappa = 0.05$ (a float diagnostic, `margin_distribution.py`), the mean gap above the demanded margin compresses from 0.70 at $\alpha = \frac{1}{2}\alpha_\star$ to 0.59 at $\alpha_\star$: solutions are squeezed against the constraint boundary as storage saturates, which is the quantity a margin-type generalization argument would consume, and a natural next target for the certified machinery. These are statements about information-theoretic storage, not about algorithms: for the Ising perceptron the known efficient algorithms stall far below $\alpha_\star$, and the margin curve gives the ceiling against which that algorithmic gap should be measured at every κ. Limits of the reading: nothing here concerns generalization on structured data, correlated features, or hidden layers; κ is a feature-space radius, and translating it to input space requires a Lipschitz bound on the feature map.

The algorithmic gap the curve frames is directly measurable at moderate size. A population annealer (hinge energy on violated margins, geometric schedule matched to the $2/\sqrt{N}$ energy scale; `finite_size_experiment.py`) trained $N = 101$ instances across α at $\kappa \in \{0, 0.05, 0.1\}$. Its success walls (the half-success crossings) fall in the correct κ order and sit below the corresponding $\alpha_\star(\kappa)$ by 0.08–0.12 in α at this budget, while scattered successes persist to the bound itself: at $\kappa = 0.1$ the last appear at $\alpha \approx 0.73$, essentially at $\alpha_\star$, and none beyond. The solutions the annealer does find are atypical in exactly the direction the clustering picture predicts: at low load their mean gap above κ tracks the typical-solution law $p(h)$, while near the wall it holds around 0.65–0.70 as the typical mean descends — the algorithm reaches the wide, over-margined minority. At $N = 201$ under the same per-spin budget the walls sharpen and shift left, so what the annealer traces is a budget-dependent lower envelope of the true algorithmic threshold, whose location and the clustering barrier behind it are the subject of a substantial literature [10, 13] on the closely related symmetric and asymmetric binary perceptrons; the found-solution histograms meanwhile track the derived $p(h)$ closely at moderate load and deform toward mid-gaps near the wall. Both effects are schedule-dependent observations, not thresholds; the certified curve is what they are measured against, and the comparison compresses neatly: within each size and budget the wall sits at a κ-independent fraction of $\alpha_\star(\kappa)$ (0.90 at $N = 101$, 0.75 at $N = 201$, 0.69 at $N = 301$, flat across margins to a few percent), so the demanded margin costs the algorithm the same relative capacity it costs the storage bound (Figure 3). At fixed sweeps per spin the fraction falls with N, so holding a constant fraction of capacity requires superlinearly growing total work; at $N = 201$ the fraction climbs 0.75, 0.79, 0.86 as the budget quadruples twice, consistent with a fixed gain per quadrupling over the measured range.

The margin price also depends on the weight alphabet. Solving the replica-symmetric system for ternary weights $\{-1, 0, +1\}$ with a self-overlap order parameter (`exploration/ternary_km.py`, anchored by reproducing the binary $\alpha_\star(0) = 0.8330786$ to the certified digits) gives $\alpha_\star^{\mathrm{tern}}(0) = 1.17448$ against an annealed bound of $\log_2 3$, with 1.07951 at $\kappa = 0.05$ and 0.99483 at $\kappa = 0.1$. Per weight *state* the richer alphabet stores more; per weight *bit* it stores less — 0.741 against the binary 0.833 constraints per bit at $\kappa = 0$, and the gap widens monotonically with margin (0.100 at $\kappa = 0.05$, 0.105 at 0.1). At this level of the theory, binary remains the bit-efficient quantization for robust storage. The full eight-point curve (Figure 4, κ up to 0.35) makes this a statement about the whole margin range rather than three samples: per stored bit the binary curve sits above the ternary one everywhere, the absolute gap saturating near 0.11 constraints per bit above $\kappa \approx 0.2$ while the relative advantage grows from 12 to 26 percent, since both capacities fall with margin.

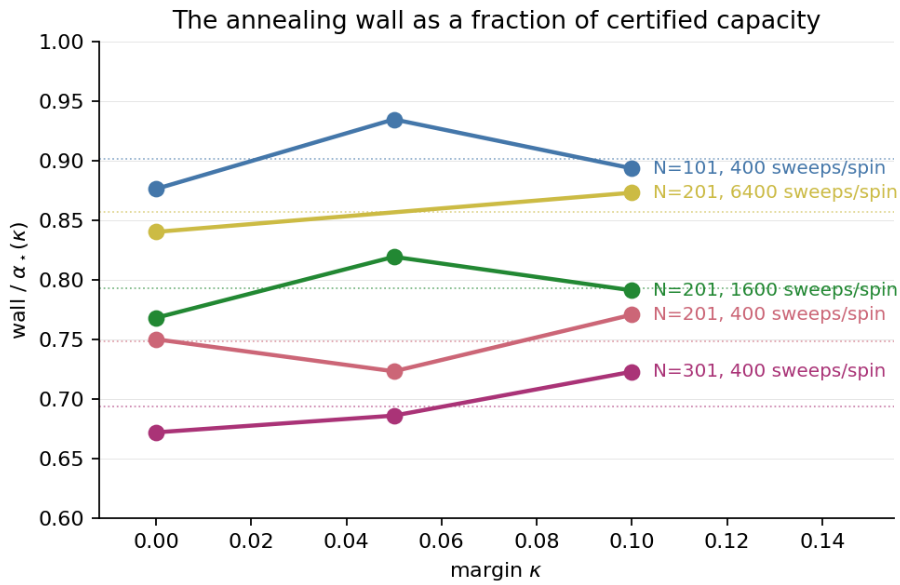

Figure 3: The annealing wall as a fraction of the certified capacity $\alpha_\star(\kappa)$, across size–budget pairs. Within each run the fraction is flat in κ to a few percent, while its level depends on both size and budget.

13 Extensions

The machinery is not specific to the binary prior or to this constraint law, and the natural next studies are ordered by how much of it they reuse.

The strip, quantified. The four-piece certificate of Section 8 runs at a κ ball exactly as at a point; the located boxes widen with the slab and the wedge-edge margins (about 10^{-3}) set the affordable slab width. An adaptive κ-subdivision driven by the certified margins would turn Theorem 1 into a statement on the whole strip. This is engineering plus compute, not new mathematics.

Other margins of the same family. The certified pipeline applies wherever the continuation reaches: positive margins to $\kappa \approx 1.15$ (past which the fixed-tilt center certificate must re-optimize the tilt, as measured in Section 6), and negative margins until replica-symmetry breaking territory, where the honest move is to certify the approach (the degradation of the weak eigenvalue and of the λ_0 margin) rather than the condition.

Other priors and activations. The moment-coordinate reduction needs only that the profile enter through finitely many moments of a log-concave base pair; replacing $\{-1, +1\}$ by a bounded discrete prior changes Φ and the moment body but not the architecture. The spherical perceptron at margin (Gardner's curve, shown for context in Figure 1) has simpler convexity structure and would serve as a validation target where exact answers are classical.

Teacher–student and structured labels. The planted version replaces the annealed constraint law by a tilted one; the cavity argument and the two-variable function have direct analogues, and the certified-slab design (per-slab rectangles, located boxes, deep bisection) transfers whenever the fixed-point system does.

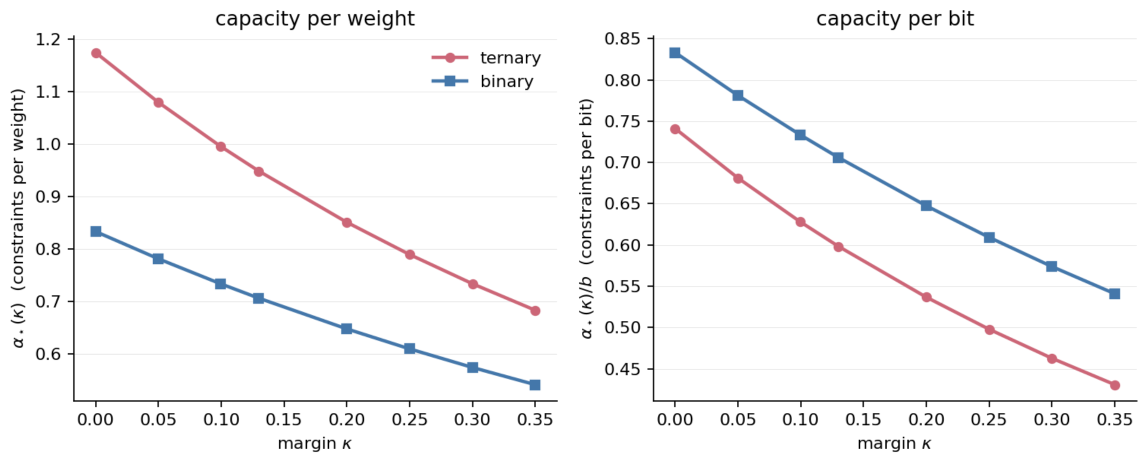

Figure 4: Replica-symmetric capacity of ternary against binary weights, per weight (left) and per stored bit (right, dividing by $\log_2 3$ and 1). The order reverses: the richer prior stores more per weight and less per bit, at every computed margin.

Toward learning theory. Two quantitative bridges seem within reach of exactly this machinery: certified capacity curves for one-bit compressed classifiers (the κ-robust memorization ceiling above, made uniform over the strip), and certified bounds on the margin distribution achievable at a given $\alpha < \alpha_\star(\kappa)$, which is the object that connects storage geometry to generalization bounds of the classical margin type.

14 Scope

The certified statements are: Conditions 1–3 on $\kappa \in [-0.65, 0.10]$, the same without the rectangle clause on $(0.10, 0.19]$, and the rectangle clause again at $\kappa \in \{0.13, 0.185\}$ (Theorem 2); deep $\alpha_\star$ intervals at five margins; the center Hessian at six; Condition 4 at $\kappa \in \{-0.45, -0.05, 0, 0.05, 0.0995, 0.13\}$; and hence Huang's unconditional upper bound at the five nonzero margins (Theorem 1), with the rest of the strip conditional only on Condition 4 at the margin in question. The lower bound is untouched: Ding–Sun's conditioned second-moment reduction is proved at $\kappa = 0$ only, so nothing here is an exact threshold statement at $\kappa \neq 0$. All float scans are diagnostics, not certificates, and are labeled as such in the repository.

Declarations

Most of the work reported here was done by Fable 5 (Anthropic), an AI system: the mathematical theory (the stationarity and derivative identities, the general-margin Hessian formulas, the symbolic- maximizer and correlated-Bregman devices, the stability bounds), the software, the certificates, and the drafting of this manuscript. The system operated in an autonomous goal mode: the author set the objective of extending the verified capacity results to nonzero margin, and it worked toward that objective over extended unattended sessions, choosing the approach, implementing and running the certificates, detecting and repairing its own errors through independent cross-checks (several such repairs are documented, with the failed artifacts preserved, in the repository), and iterating until the verifications closed. The author supervised the runs, reviewed the mathematics, and accepts responsibility for the final manuscript. This is part of a broader research project of the author on the extent to which current AI systems can resolve open problems in mathematics; the methodology by which these AI systems operated will be released as a separate research report in due course.

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